

NETWORKS AND COMPLEXITY

Solution 13-10

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 13.10: General solution for linear maps [Lvl. 4]

In this chapter, we derived a general solution for linear differential equations, but for linear maps we have only solved examples. Find a general solution for linear discrete-time maps, i.e. systems of the form $\mathbf{x}_{i+1} = \mathbf{U}\mathbf{x}_i$.

Solution

Any linear map can be written in the form

$$\mathbf{x}_{t+1} = \mathbf{J}\mathbf{x}_t. \quad (1)$$

Since $\mathbf{J}$ advances the time by one step, we can also write

$$\mathbf{x}_t = \mathbf{J}^t \mathbf{x}_0 \quad (2)$$

We now decompose $\mathbf{x}_0$ into eigenvectors of $\mathbf{J}$, that is, we determine c_n such that

$$\mathbf{x}_0 = \sum c_n \mathbf{v}_n \quad (3)$$

where

$$\mathbf{J}\mathbf{v}_n = \lambda_n \mathbf{v}_n. \quad (4)$$

The time dependent solution is now

$$\mathbf{x}_t = \mathbf{J}^t \mathbf{x}_0 \quad (5)$$

$$= \mathbf{J}^t \sum c_n \mathbf{v}_n \quad (6)$$

$$= \sum c_n \mathbf{J}^t \mathbf{v}_n \quad (7)$$

$$= \sum c_n \lambda_n^t \mathbf{v}_n \quad (8)$$

$$(9)$$

So the general solution for linear maps is

$$\mathbf{x}_t = \sum c_n \lambda_n^t \mathbf{v}_n. \quad (10)$$